Neel Gali

GIS Analyst I

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | Data Management Platform

- Designed and implemented scalable **Python** scripts for automated **geospatial data acquisition** from diverse sources, boosting data freshness by 35% weekly.
- Led development of a **web-based application** using **React** and **Leaflet.js** to visualize complex **GIS data**, enhancing client access to mapping by 40%.
- Optimized **PostgreSQL** with **PostGIS** extensions for efficient storage and query of large **enterprise databases**, improving spatial analysis speeds by 25% for users.
- Developed custom **geoprocessing tools** using **ArcPy** within **ArcGIS Pro** to automate complex spatial analysis workflows, reducing manual processing time by 50% for projects.
- Implemented **CI/CD pipelines** using **Git** for **GIS software** deployments on **AWS**, ensuring consistent and reliable updates to data management infrastructure 99% of the time.
- Conducted rigorous **quality assurance/quality control** of incoming **LiDAR-derived data** and vector datasets, ensuring data integrity for critical land surveys 95% accurately.
- Integrated **remote sensing data/techniques** into **Python** workflows to enhance environmental impact assessments, increasing predictive accuracy for site planning by 15%.

Graduate Research Assistant *Jul 2024 – Jun 2025*

Oklahoma State University | Boone Pickens School of Geology

- Developed **Python** scripts for automating routine **geospatial data acquisition** and cleaning tasks, reducing manual effort by 30% for quarterly updates.
- Created interactive **web-based applications** using **JavaScript** and **D3.js** for visualizing complex geographic datasets, improving user engagement by 20% quarterly.
- Managed and maintained **geospatial databases** within **PostgreSQL**, ensuring high availability and optimal performance for daily **spatial analysis** operations 99.9% of the time.
- Utilized **ArcGIS Desktop** to perform **geoprocessing tools** operations and complex **data analysis**, generating precise maps and reports for 10+ ongoing projects.
- Performed extensive **data conversion** and validation of diverse geographic formats, enforcing strict **data integrity** standards for all incoming project data 100%.
- Assisted in developing and documenting standardized **GIS procedures** and **mapping techniques** to streamline project workflows, reducing onboarding time for new analysts by 15%.
- Supported the integration of **AutoCAD** drawings with **GIS data** for unified project views, enhancing cross-departmental collaboration for 5 key initiatives.
- Optimized data processing pipelines using **Pandas** and **NumPy** for large **Microsoft Excel** datasets, reducing processing time by 40% for reporting.

Software Engineer *Jul 2021 – Aug 2022*

Zemoso Technologies | Data Infrastructure & Web Mapping

- Developed and maintained **Python** scripts for efficient data extraction and transformation (**ETL**) from various sources, processing 500GB of raw data monthly with 98% accuracy.
- Built and deployed RESTful **APIs** using **Django** to serve cleaned data to front-end **web-based applications**, improving data accessibility for 100+ internal users.
- Contributed to the design and management of relational **databases** like **MySQL**, ensuring data integrity and optimizing query performance by 20% for reporting.
- Implemented front-end interfaces using **HTML**, **CSS**, and **JavaScript** for data visualization dashboards, enhancing user understanding of key metrics by 15%.
- Assisted with **quality assurance** testing and debugging of software features, ensuring 99% defect-free releases for critical data management modules.
- Utilized **Microsoft Excel** for preliminary **data analysis** and reporting, preparing summarized datasets for further processing by 20 stakeholders.

Skills

Languages: Python, JavaScript, SQL, HTML, CSS
Frameworks & Tools: React, Leaflet.js, ArcPy, ArcGIS Pro, D3.js, Pandas, NumPy, AutoCAD, Django, Git
Cloud & Infra: AWS, Cloud-based Infrastructure, PostgreSQL, PostGIS, MySQL, Microsoft Office Suite
Concepts: Geospatial Data Acquisition, Spatial Analysis, Geoprocessing Tools, Geospatial Databases, Data Integrity, Mapping Techniques, GIS Procedures, Remote Sensing Data/Techniques, LiDAR-derived Data, CI/CD, Data Visualization, APIs, ETL

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India